Year 13 Further Maths - Normal distribution

Similar questions - Confidence intervals and hypothesis tests for a mean

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The greatest weight W N that can be supported by a shelving bracket of traditional design is normally distributed with mean 500 and standard deviation 80. A sample of 40 brackets of a new design is tested and the mean of the greatest weights they can support is 468.0 N.

(a) Test at the 1% significance level whether the mean greatest weight that a bracket of the new design can support is less than that for the traditional design. **[7]**

(b) State an assumption needed in part (a). **[1]**

(c) Explain whether it is necessary to use the Central Limit Theorem in part (a). **[1]**

2 A company uses two drivers. Driver A charges a fixed £80 per day plus £2 per mile. Driver B charges a fixed £120 per day plus £1.50 per mile. On each working day the distance travelled by each driver is a random variable with distribution $\mathrm{N}(83, 360)$.

(a) Find the probability that driver A charges less than £235 for a randomly chosen day. **[4]**

(b) Find the probability that the total charge for three randomly chosen days of driver A is at least £300 more than the total charge for two randomly chosen days of driver B. **[6]**

3 The continuous random variable X has the distribution $\mathrm{N}(\mu, 30)$. The mean of a random sample of 8 observations of X is 53.1. Determine a 95% confidence interval for μ, giving the end points correct to 4 s.f. **[4]**

4 A random sample of 160 observations of a random variable X is summarised by $\sum x = 2688$ and $\sum x^2 = 48398$.

(a) Calculate unbiased estimates of (i) $\mathrm{E}(X)$, (ii) $\mathrm{Var}(X)$. **[4]**

(b) Find a 99% confidence interval for $\mathrm{E}(X)$, giving the end points to 4 s.f. **[3]**

(c) Explain whether it was necessary to use the Central Limit Theorem in answering (i) part (a), (ii) part (b). **[2]**

5 The random variable X was assumed to have a normal distribution with mean μ. A significance test using a random sample of size 128 was carried out with H_0: $\mu = 30$ and H_1: $\mu > 30$. It was found that $\sum x = 3929.6$ and $\sum x^2 = 123483.52$. The conclusion was to reject the null hypothesis.

(a) Determine the range of possible values of the significance level of the test. **[5]**

(b) It was later found that X was not normally distributed. Explain whether this invalidates the conclusion. **[2]**

6 A club secretary records the time T minutes needed to process a new member's details; $n = 40$, $\sum t = 396.0$, $\sum t^2 = 4271.40$.

(a) Determine a 99% confidence interval for the mean μ. **[7]**

(b) The secretary discovers that the true variance over a long period is $\sigma^2 = 10.0$ and takes a new independent sample of 50 observations, constructing a 99% confidence interval for μ using $\sigma^2 = 10.0$. Find the probability that this new interval contains the value $\mu + 1.6$. **[3]**

7 Research suggests that the mean reading age of a child about to start secondary school is 10.75 years. The reading ages X of a random sample of 80 children about to start secondary school in a district were summarised as $\sum x = 900$ and $\sum x^2 = 10267$.

(a) Test at the 5% significance level whether the mean reading age in this district is not 10.75. **[10]**

(b) A student wrote: "Although we do not know that the distribution of X is normal, the central limit theorem allows us to assume that it is, as the sample size is large." Give a corrected version of this statement. **[1]**

Question 1

(a) H_0: $\mu = 500$; H_1: $\mu < 500$. Under H_0, $\bar{W} \sim N\left(500, \frac{80^2}{40}\right)$. $z = \dfrac{468-500}{80/\sqrt{40}} = -2.53$. Critical value -2.326; $-2.53 < -2.326$ so reject H_0: there is evidence that the new brackets support a lower mean weight
(b) The standard deviation for the new design is also 80 N (and the weights are independent)
(c) No: the traditional distribution is normal, and if the new design's weights are also normal then $\bar{W}$ is exactly normal, so the CLT is not needed

Question 2

(a) $A = 80+2M$, $A \sim N(246, 4\times 360 = 1440)$. $P(A<235) = P\left(Z < \dfrac{-11}{37.95}\right) = P(Z<-0.29) = 0.386$
(b) $B = 120+1.5M \sim N(244.5, 810)$. $D = A_1+A_2+A_3-B_1-B_2$: mean $= 738-489 = 249$, variance $= 3(1440)+2(810) = 5940$ (sd 77.07). $P(D \geqslant 300) = P\left(Z \geqslant \dfrac{51}{77.07}\right) = P(Z \geqslant 0.662) = 0.254$

Question 3

$53.1 \pm 1.96\sqrt{\frac{30}{8}} = 53.1 \pm 3.795$: (49.30, 56.90)

Question 4

(a)(i) $\bar{x} = \frac{2688}{160} = 16.8$ (ii) $s^2 = \dfrac{48398 - 160(16.8)^2}{159} = \dfrac{3240}{159} = 20.375$
(b) $16.8 \pm 2.576\sqrt{\frac{20.375}{160}}$: (15.881, 17.719) i.e. (15.88, 17.72)
(c)(i) No: unbiased estimates do not depend on the distribution
(ii) Yes: the distribution of X is unknown, but $n = 160$ is large so the CLT gives that $\bar{X}$ is approximately normal

Question 5

(a) $\bar{x} = 30.7$, $s^2 = \dfrac{123483.52 - 128(30.7)^2}{127} = 22.4000$. $z = \dfrac{30.7-30}{\sqrt{22.4000/128}} = 1.673$. $P(Z > 1.673) = 0.0471$. The null hypothesis is rejected, so the significance level is at least 0.0471 (i.e. greater than about 4.71%) and at most 100% (any level $\alpha > 0.0471$)
(b) No: $n = 128$ is large so by the CLT $\bar{X}$ is approximately normal and the test remains valid

Question 6

(a) $\bar{t} = 9.9$, $s^2 = \dfrac{4271.4 - 40(9.9)^2}{39} = 9.0$. $9.9 \pm 2.576\dfrac{3}{\sqrt{40}} = 9.9 \pm 1.222$: (8.68, 11.12)
(b) The interval is $\bar{X} \pm 2.576\sqrt{\frac{10}{50}} = \bar{X} \pm 1.152$ with $\bar{X} \sim N(\mu, 0.2)$. It contains $\mu + 1.6$ if $0.448 \leqslant \bar{X} - \mu \leqslant 2.752$: $P(1.002 \leqslant Z \leqslant 6.15) = 0.158$

Question 7

(a) $\bar{x} = 11.25$, $s^2 = \dfrac{10267 - 80(11.25)^2}{79} = 1.7975$. H_0: $\mu = 10.75$; H_1: $\mu \neq 10.75$. By the CLT, $\bar{X} \approx N\left(10.75, \frac{1.7975}{80}\right)$; $z = \dfrac{11.25-10.75}{\sqrt{1.7975/80}} = 3.336$. Critical values ± 1.96; $3.336 > 1.96$ so reject H_0: evidence the mean reading age differs from 10.75
(b) "Although the distribution of X is not known to be normal, the sample is large so by the central limit theorem the distribution of the sample *mean* is approximately normal"